TUTORIAL · ABOUT 20 MINUTES

Preparing a metalloprotein with DYFF

From a raw PDB file to a solvated metalloprotein ready for simulation. The main example is the zinc finger Xfin-31 (PDB 1ZNF), a structural $Zn(Cys)_2(His)_2$ site. A second example, rubredoxin (PDB 1IRO), shows an open-shell iron site.

> **You need:** EasyHybrid 3.2.0 and xTB in your `PATH` (see Setting up xTB). The structures are on the RCSB PDB: 1ZNF and 1IRO; both are also in `tests/data/metalloproteins/` of the EasyHybrid folder.

Why metals need care

Standard protein force fields (AMBER, CHARMM, OPLS-AA) have no parameters for transition metals in their protein sites. Two models are common:

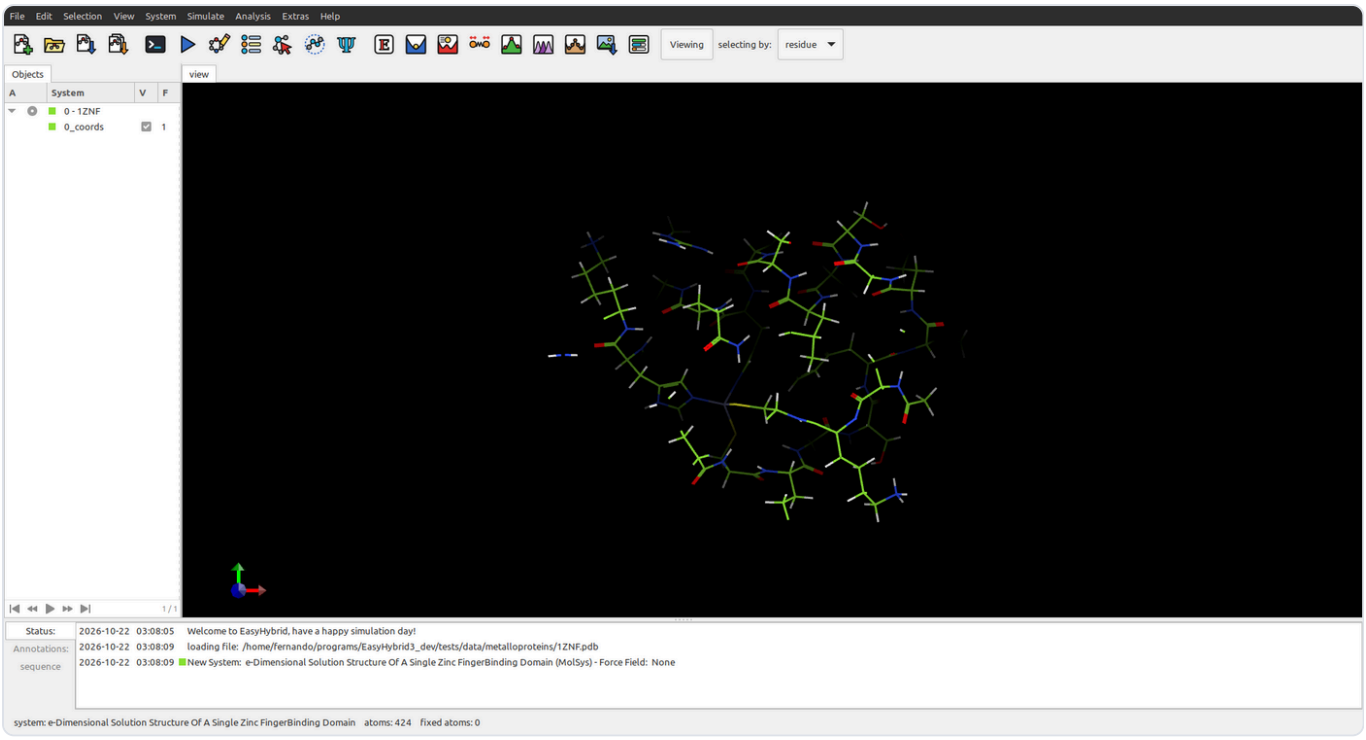
- **Ionic (non-bonded):** the metal is a point charge with a Lennard-Jones sphere. It works for hard ions such as Mg^{2+} and Ca^{2+}, but a Zn^{2+} or Cu^{2+} held this way drifts, changes its coordination number and loses the geometry of the site.
- **Bonded:** the metal is bonded to its donor atoms, with bond, angle and torsion terms of their own and charges that describe the charge transfer to the ligands. It keeps the geometry of the site, but each site needs its own parameters.

Prepare DYFF System builds the bonded model automatically. It finds the metal and its donors by distance, gives the metal a DYFF type for its charge and coordination (for example `Zn:Tet:+2`), and refines the metal with its donor residues together with xTB: CM5 charges, bonds and angles from the xTB Hessian, and torsions from the DFT library. The standard residues away from the metal take their parameters from the residue library.

1 Open the structure

Use **File** › **Open** (or the **Open** button) and choose `1ZNF.pdb` (in `examples/tutorials/09_metalloprotein/` of the EasyHybrid folder).

- 1ZNF is an NMR ensemble with 37 models. EasyHybrid reads the first one.
- The file has no CONECT records for the protein, so it is read without bond orders. The preparation in step 3 builds the chemistry: hydrogens, bond orders and protonation states.



1ZNF as read from the PDB file: 424 atoms.

The system name comes from the TITLE record and is long. Give it a short name with **System › Rename**, for example `1ZNF`: the prepared system will be called `1ZNF_DYFF`.

2 Look at the metal site first

Open the Terminal (toolbar button **Terminal**) and type `metals`. It lists what the preparation will find, without changing anything:

```
        ------------------------------------------------------------
               Welcome to the EasyHybrid Terminal.
        ------------------------------------------------------------

                    Created by J.F.R Bachega

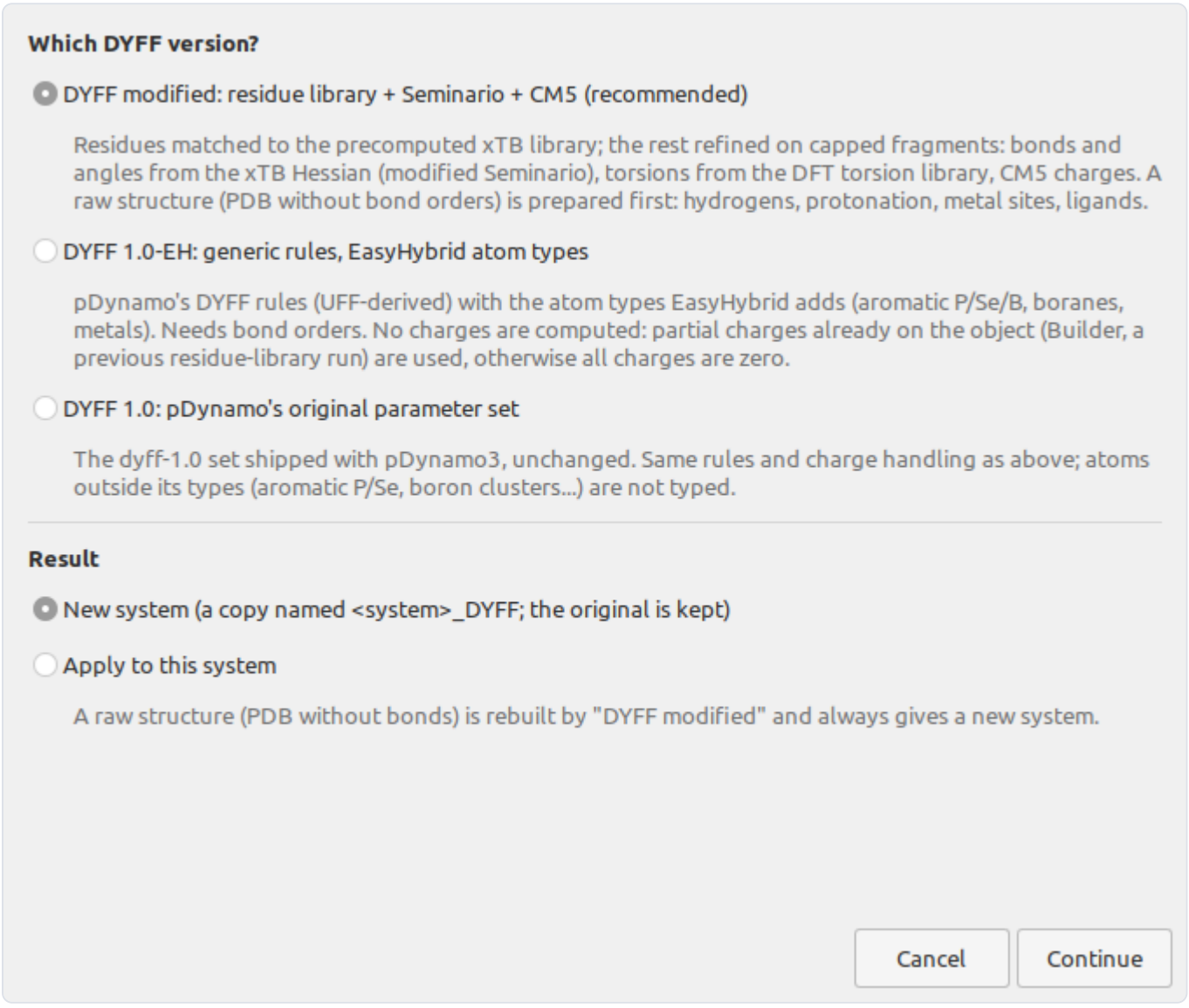

        >>> rename name=1ZNF
Renamed: 'e-Dimensional Solution Structure Of A Single Zinc FingerBinding Domain' ->
'1ZNF'
>>> systems
      id  atoms  force field     QC   tag         name
  *    0    424  -               -    MolSys      1ZNF
>>> metals
[0] 1ZNF: 1 metal site(s).
  A:ZN.27      Zn +2  mult 1  bonded  donors: HIS.19 NE2 1.95, HIS.23 NE2 1.99, CYS.3 SG
2.29, CYS.6 SG 2.31
  Covalently bound, removed by 'dyff': A:ACE.0, A:NH2.26
```

- One site, `A:ZN.27` : Zn^{2+}, multiplicity 1, bonded model.
- Four donors: NE2 of His19 (1.95 Å) and His23 (1.99 Å), SG of Cys3 (2.29 Å) and Cys6 (2.31 Å). This is the expected tetrahedral Cys_2His_2 site.
- Two peptide caps, ACE.0 and NH2.26, bonded to the chain. They will be removed.

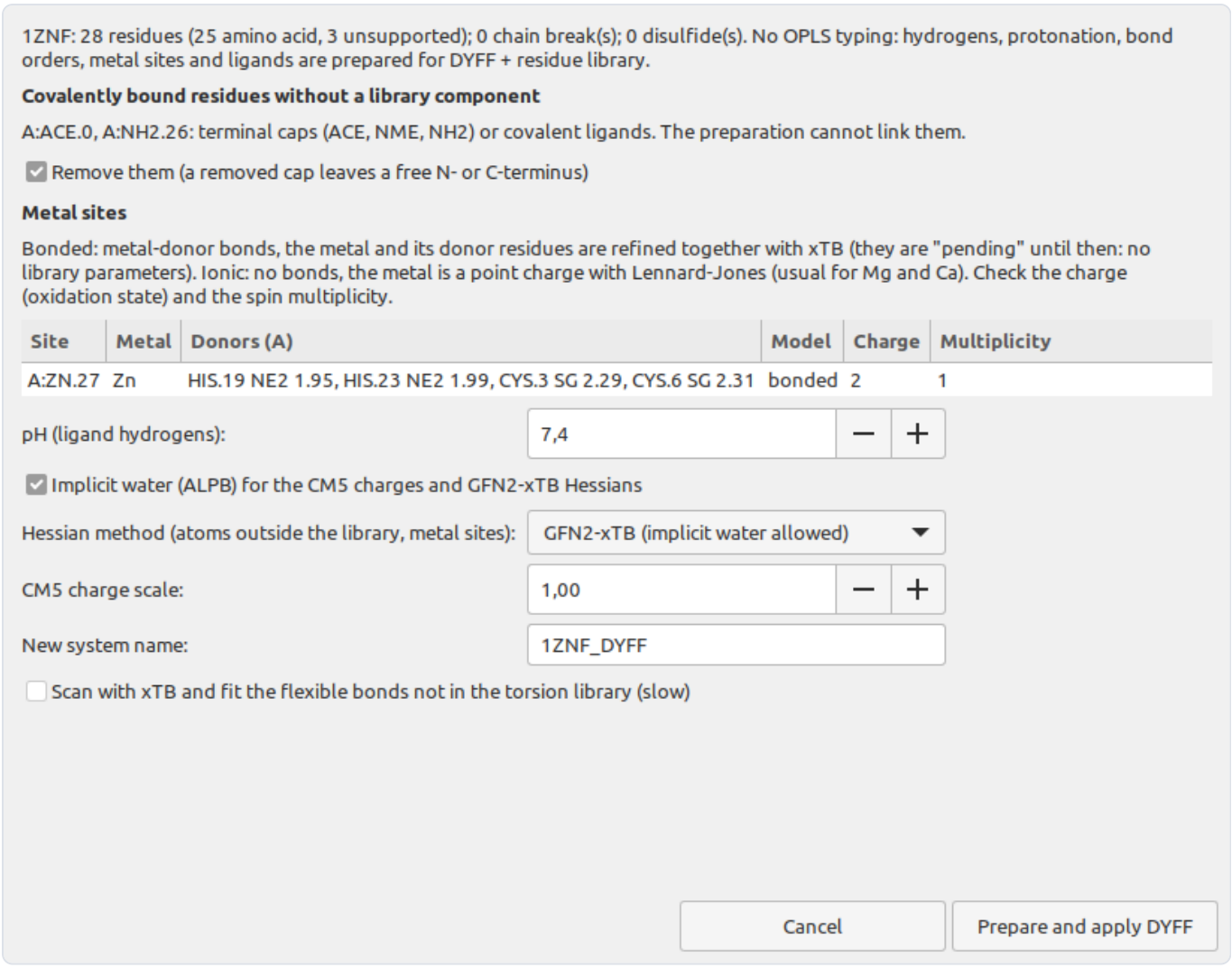
> **Tip:** check the donors against the literature before preparing. A donor missing from the list means the site is incomplete in the file, for example a site completed by crystal symmetry. A site with fewer than two donors is treated as ionic, with a warning.

3 Start Prepare DYFF System

Choose **System** › **Force Field** › **Define Force Field** › **Prepare DYFF System…** . In the first dialog keep **DYFF modified** and **New system**, and click **Continue**.



4 Check the preparation window



Go through the window from top to bottom:

1. **Summary:** 28 residues: 25 amino acids and 3 that are not (the two caps and the zinc).
2. **Covalently bound residues:** keep *Remove them*. The caps cannot be linked to the chain by the preparation; removing them leaves a free N-terminus and C-terminus.
3. **Metal sites:** the table row can be edited.
   - *Model*: keep **bonded**.
   - *Charge*: the oxidation state, **2** for zinc.
   - *Multiplicity*: 2S+1, **1** for Zn^{2+} (d^{10}, closed shell).

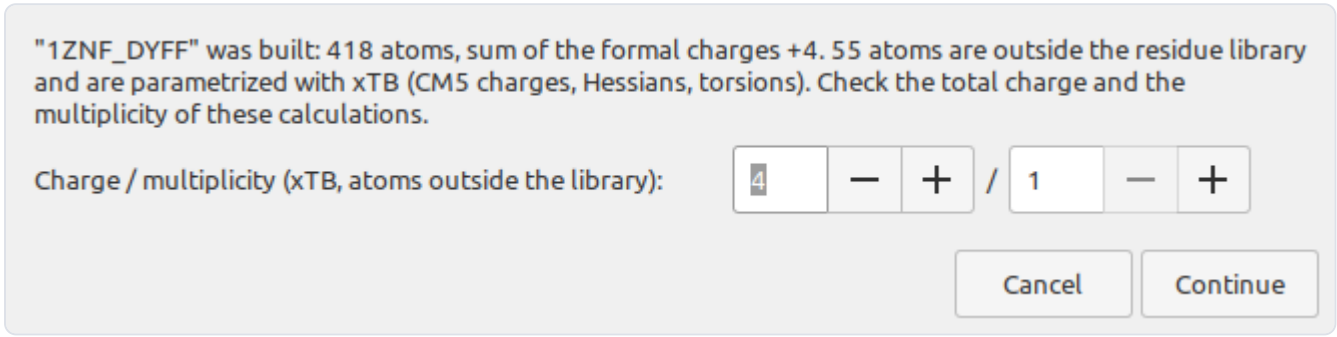
   With the bonded model, the coordinating cysteines become thiolates, and each histidine is protonated on the nitrogen that does not bind the metal.
4. **pH:** 7.4. It sets the protonation of ligands; 1ZNF has none.
5. **Implicit water (ALPB):** keep it on. The site has a formal charge, and xTB describes charged groups poorly in the gas phase.
6. **Hessian method:** GFN2-xTB. g-xTB is more accurate for some metals (it is the better choice for copper) but has no implicit solvent and is 2–3 times slower.

7. **CM5 charge scale:** 1.00.
8. **New system name:** `1ZNF_DYFF`.
9. **Scan with xTB:** leave it off. Only flexible bonds of ligands or unusual residues are scanned, and this protein has none.

Click **Prepare and apply DYFF**.

5 Confirm the charge and multiplicity

Once the hydrogens are built, the number of electrons is known, and EasyHybrid asks for the charge and multiplicity of the xTB calculations:

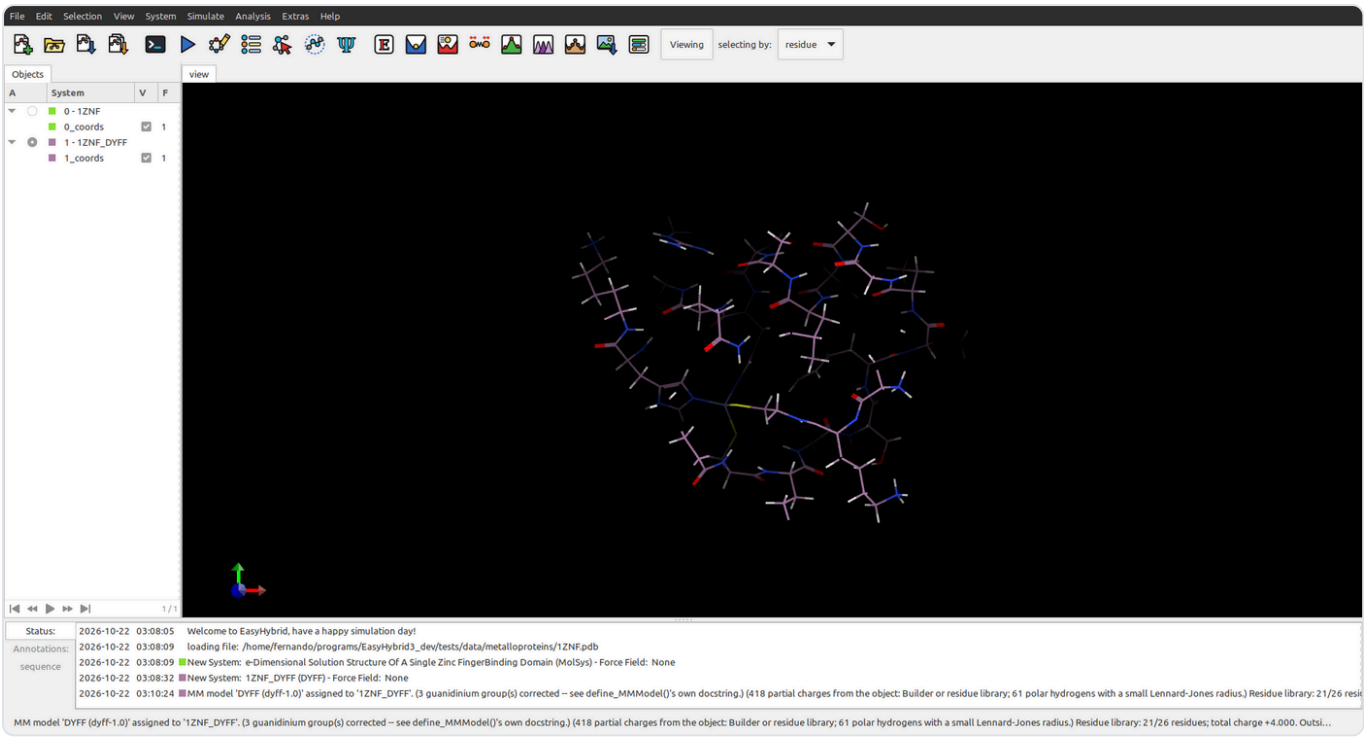


The built system has 418 atoms and a sum of formal charges of +4: Zn^{2+}, the two thiolates and the charged side chains. 55 atoms are outside the residue library: the zinc with its four donor residues and the new termini. Keep **4** and **1** and click **Continue**.

The xTB calculations run in the background, about 2 minutes for this protein. The status bar shows the progress.

6 Read the result

The new system `1ZNF_DYFF` appears in the Objects panel, and the Logs panel reports what was done:



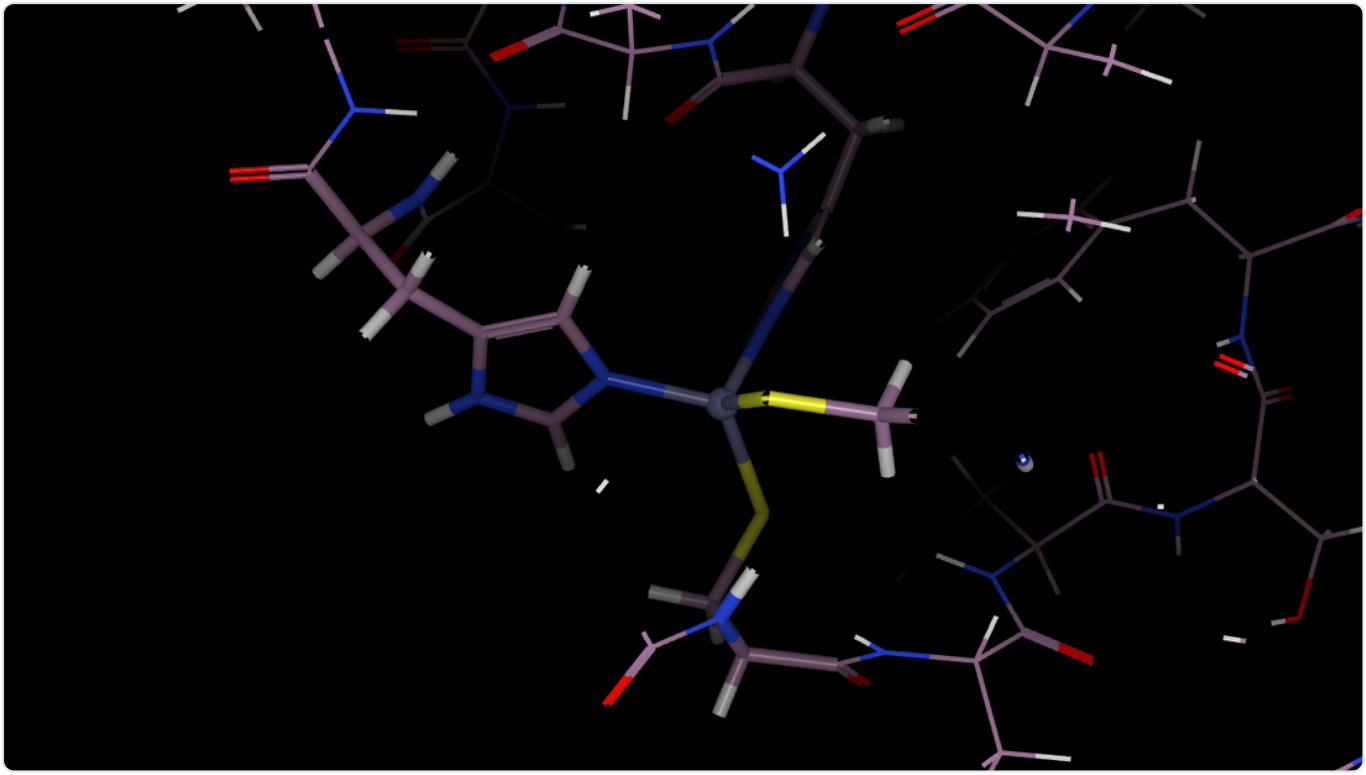
Message	Meaning
Residue library: 21/26 residues; total charge +4.000	21 residues took their parameters from the library; the other 5 (the zinc, its four donors and the termini) were refined with xTB.
Outside the library (GFN2-xTB): 56 bonds, 94 angles from the Hessian	Bonds and angles of the site from the xTB Hessian (modified Seminario).
2 of 2 rotatable bonds from the DFT torsion library	Every rotatable bond outside the library had a DFT torsion profile.
1 bonded metal site refined	The Zn site was refined as one group, with multiplicity 1.
3 guanidinium group(s) corrected	The arginine groups were given the planar DYFF types.
61 polar hydrogens with a small Lennard-Jones radius	Hydrogens on N, O and S, so that hydrogen bonds are not repulsive.
Removed: A:ACE.0, A:NH2.26	The two caps of step 4.

The original system `1ZNF` is kept, without a force field. The terminal command `energy` gives −5256.2 kJ/mol for `1ZNF_DYFF` at the NMR geometry.

7 Inspect the zinc site

Show the site in sticks and the zinc as a sphere. In the Terminal (`1` is the object of `1ZNF_DYFF` ; see `list`):

```
show rep=sticks obj=1 resi=3,6,19,23,27
show rep=spheres obj=1 resn=ZN
center obj=1 resn=ZN
```

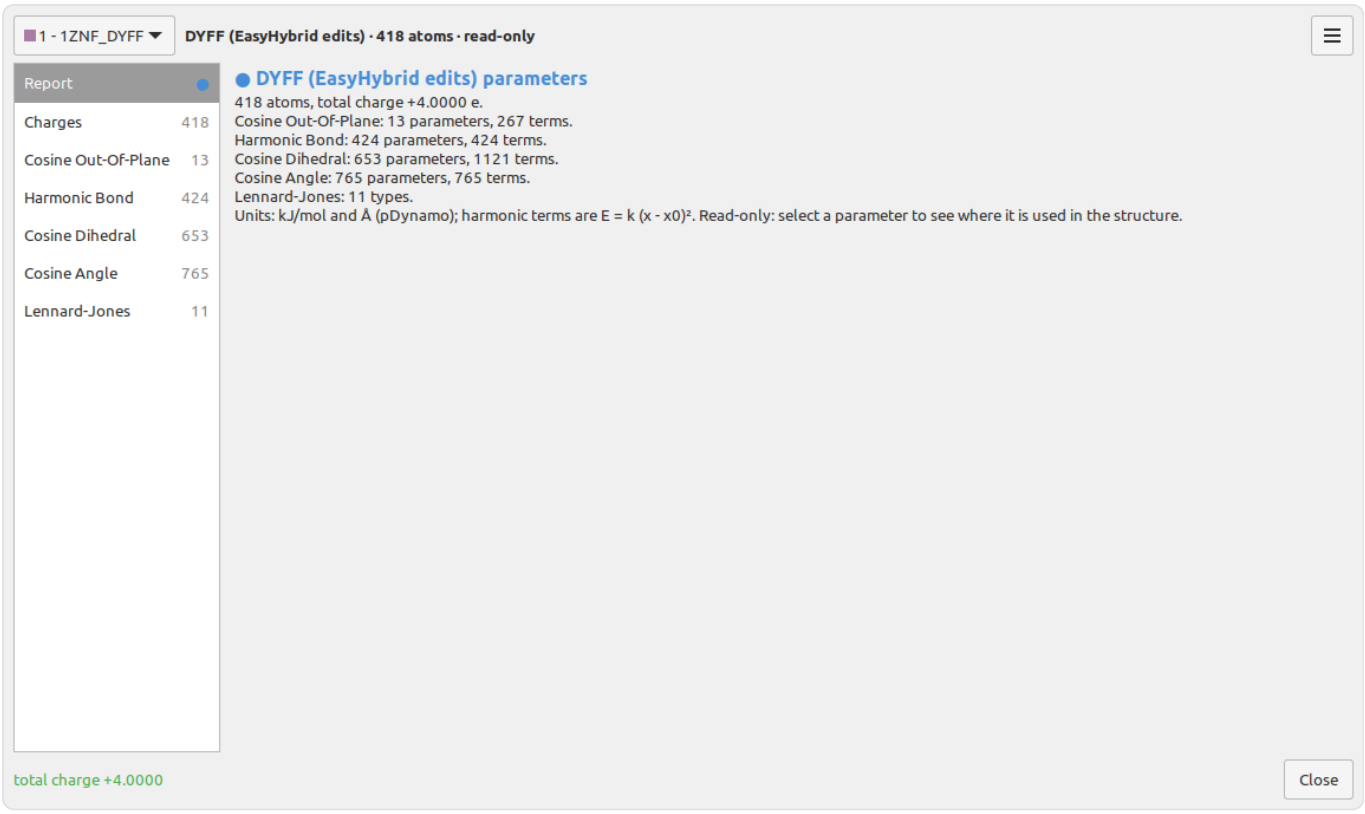


The zinc bonded to the NE2 atoms of His19 and His23 and to the SG atoms of Cys3 and Cys6.

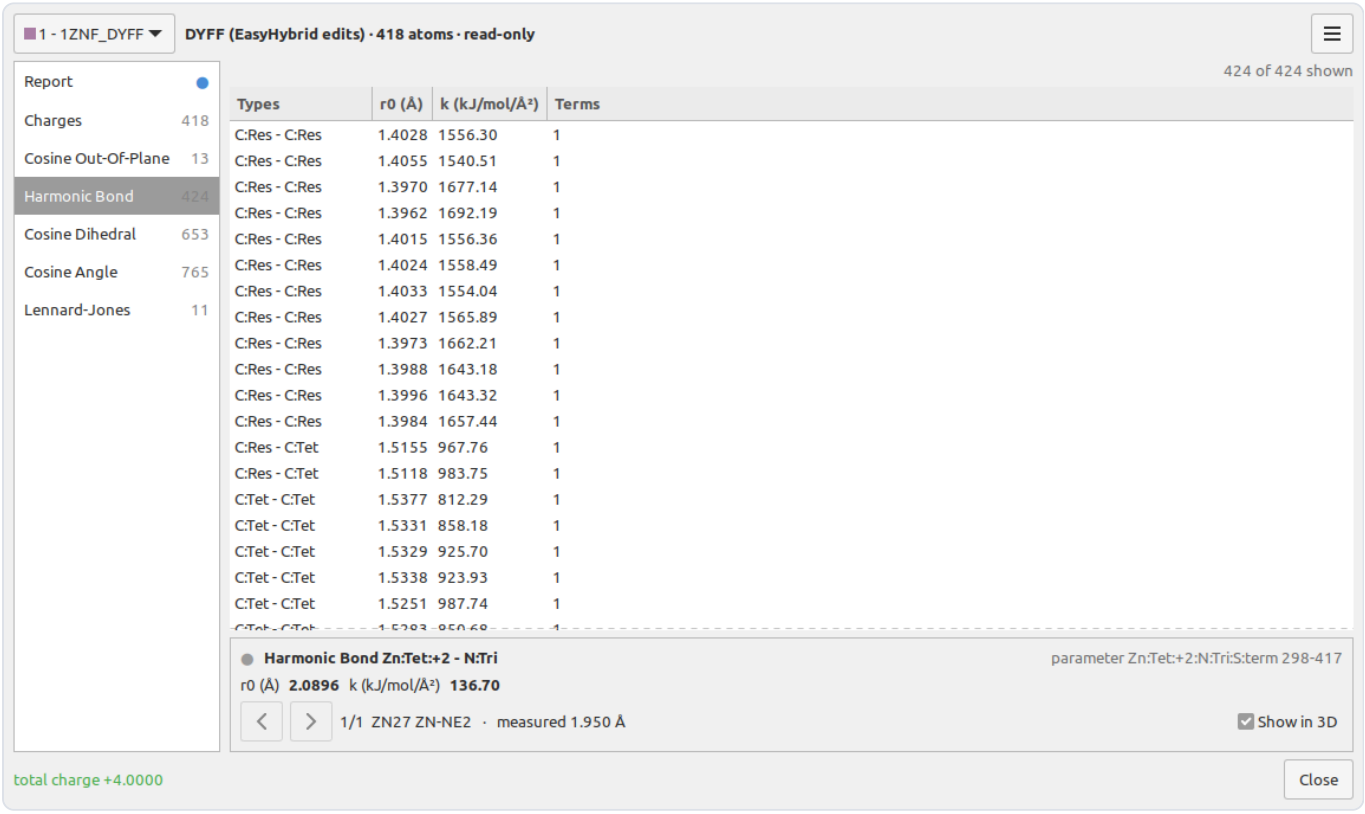
Check that the zinc has exactly four bonds, to the donors listed in step 2. A missing bond means a donor was too far away in the file; an extra one, an atom that is close but should not coordinate.

8 Check the parameters of the site

Open **System** › **Force Field** › **Force Field Parameters…**. The *Report* page summarizes the model; the other pages list every parameter.



On *Harmonic Bond*, find the zinc bonds (type `Zn:Tet:+2`) and select one. The detail pane shows the equilibrium distance and force constant from the xTB Hessian next to the distance measured in the structure, and *Show in 3D* highlights the two atoms.



Zn–NE2 of His19: r_0 = 2.090 Å, k = 136.7 kJ/mol/Å², measured 1.950 Å in the NMR model.

A difference of 0.1–0.15 Å between r_0 and the measured distance is normal: r_0 comes from the xTB minimum of the site, and the NMR model is not at that minimum. The minimization of the next step relaxes it.

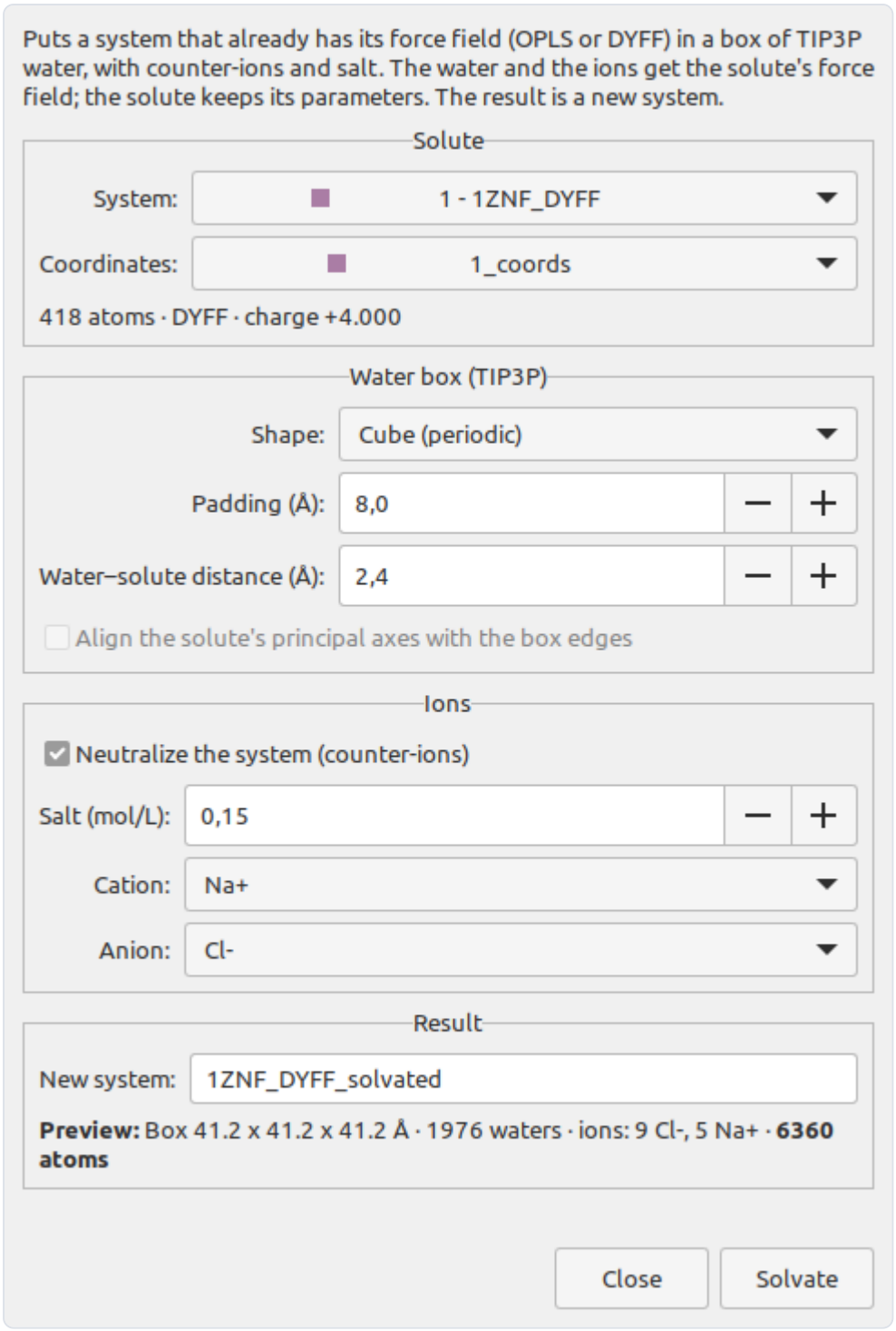
> **Note:** the metal terms are xTB values. The DFT libraries used to calibrate DYFF contain no metals, so these terms are not calibrated against DFT. Compare the minimized site with crystal structures or the literature before a production run.

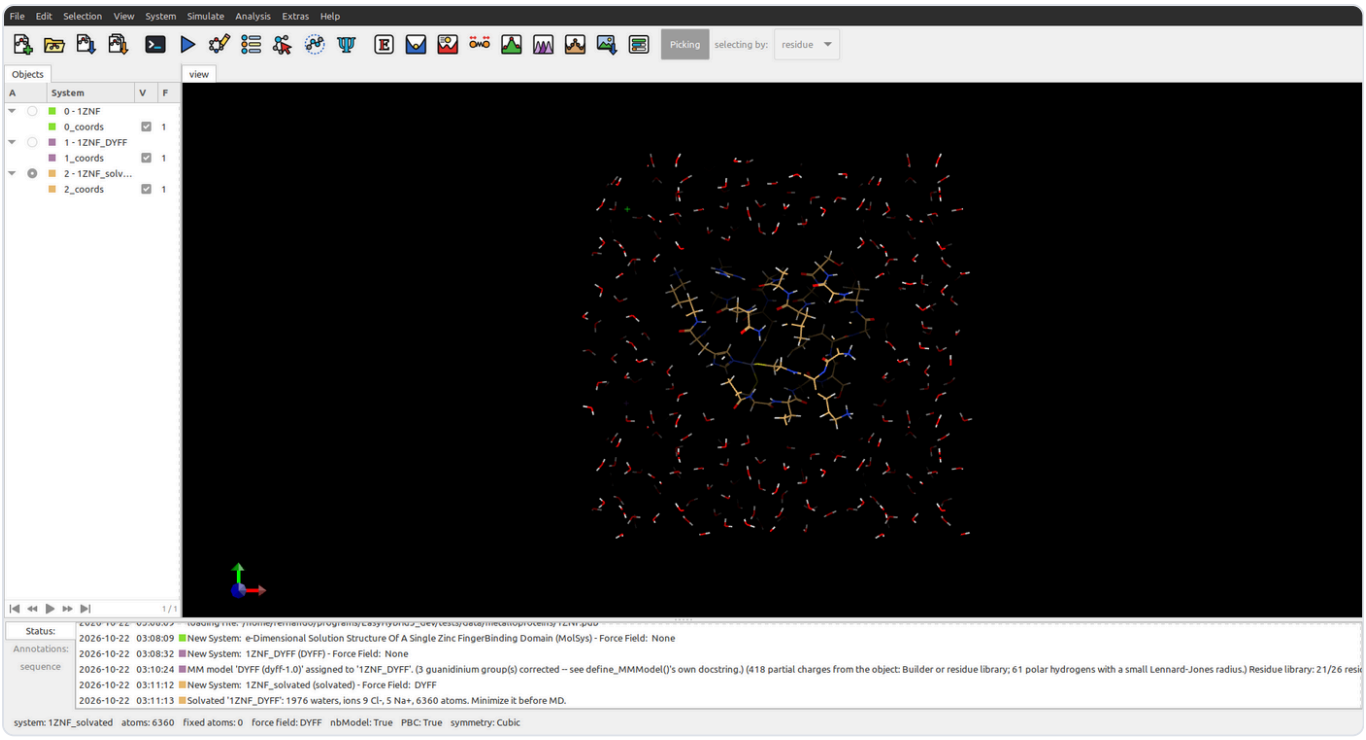
9 Minimize

With `1ZNF_DYFF` active, open **Simulate** › **Geometry optimization** and run a conjugate-gradient minimization. The hydrogens, the rebuilt termini and the site relax to the force field. Check the zinc site again after it.

10 Solvate

Open **System** › **Solvate System…**, choose `1ZNF_DYFF` (or the minimized coordinates), a cubic box with 10 Å of padding, *Neutralize* and 0.15 mol/L of NaCl, and click **Solvate**.


Puts a system that already has its force field (OPLS or DYFF) in a box of TIP3P water, with counter-ions and salt. The water and the ions get the solute's force field; the solute keeps its parameters. The result is a new system.
Solute
System:
1 - 1ZNF_DYFF
Coordinates:
1_coords
418 atoms · DYFF · charge +4.000
Water box (TIP3P)
Shape:
Cube (periodic)
Padding (Å):
8,0
Water–solute distance (Å):
2,4
Align the solute's principal axes with the box edges
Ions
Neutralize the system (counter-ions)
Salt (mol/L):
0,15
Cation:
Na+
Anion:
Cl-
Result
New system:
1ZNF_DYFF_solvated
Preview: Box 41.2 x 41.2 x 41.2 Å · 1976 waters · ions: 9 Cl-, 5 Na+ · 6360 atoms
Close
Solvate



With 8 Å of padding: 1976 waters, 9 Cl^- and 5 Na^+, 6360 atoms.

Minimize the solvated system, then equilibrate it: heat it at constant volume with the protein restrained, then at constant pressure, before production.

11 Simulate

- **In EasyHybrid:** **Simulate** › **Molecular Dynamics**, or any other simulation (scans, umbrella sampling, QC/MM with the metal site as the QC region).
- **In NAMD:** **File** › **Export As…** with the NAMD format, then **Extras** › **NAMD** › **Run NAMD…**. See Solvation and NAMD.

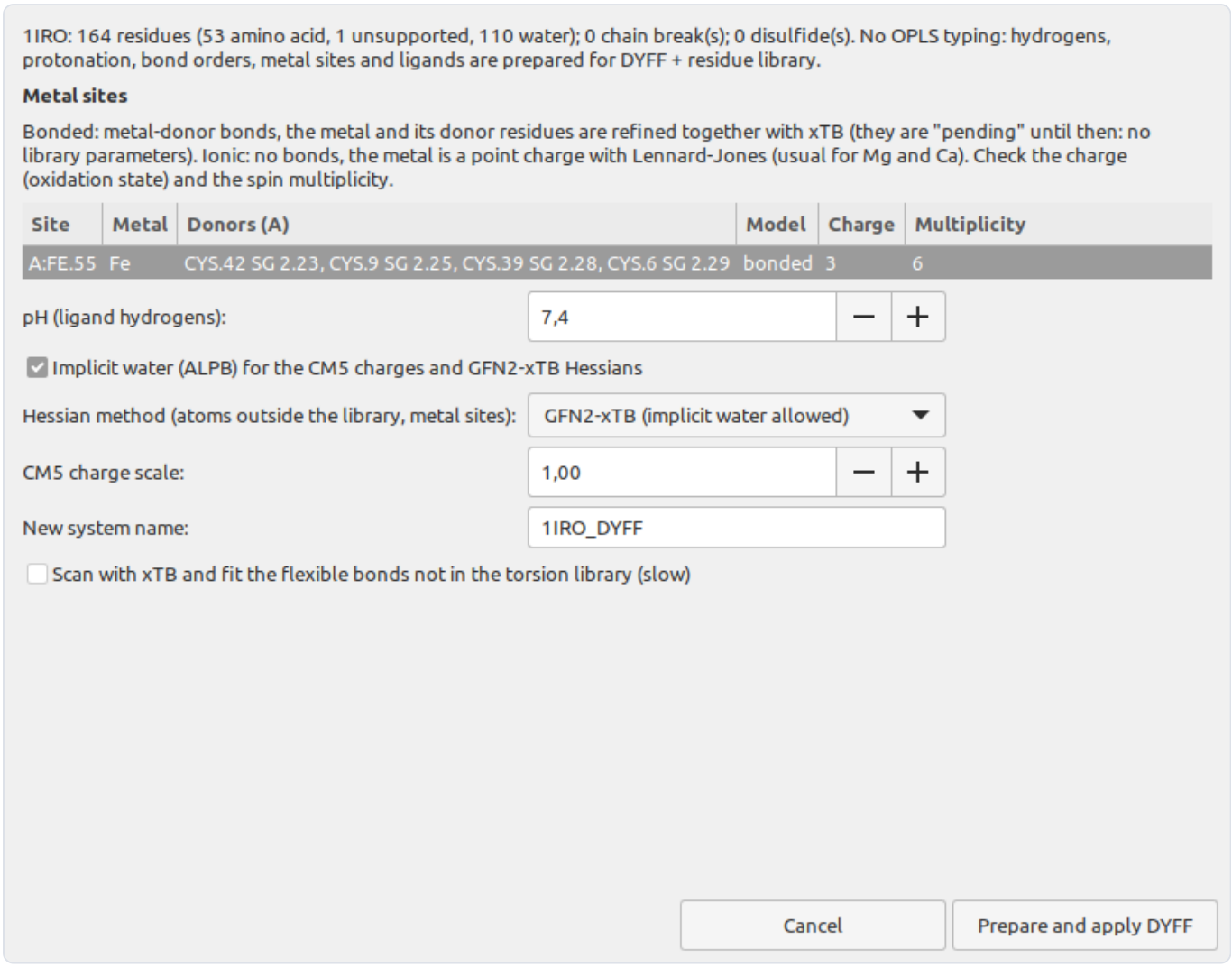
Save the session (**File** › **Save**). The `.easy` file keeps the systems, the force field with its edits and the charges.

The same preparation from the Terminal

```
load file=/path/to/1ZNF.pdb
rename name=1ZNF
metals
dyff name=1ZNF_DYFF                # defaults: modified, new system, charge +4, mult 1
systems                            # wait for 1ZNF_DYFF in the Logs panel
ffinfo sys=1
solvate sys=1 padding=10 conc=0.15 name=1ZNF_box
namd sys=2 prefix=/path/to/md/1znf_box
```

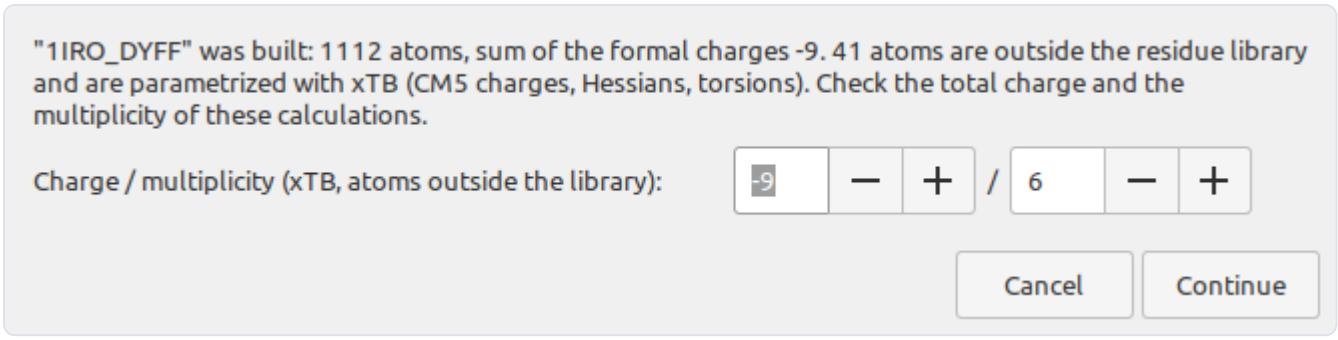
Second example: rubredoxin, an open-shell iron site

Rubredoxin from *Clostridium pasteurianum* (1IRO, 1.1 Å) has one iron bound to four cysteines. The file is the oxidized form, Fe(III), and contains 110 crystal waters.

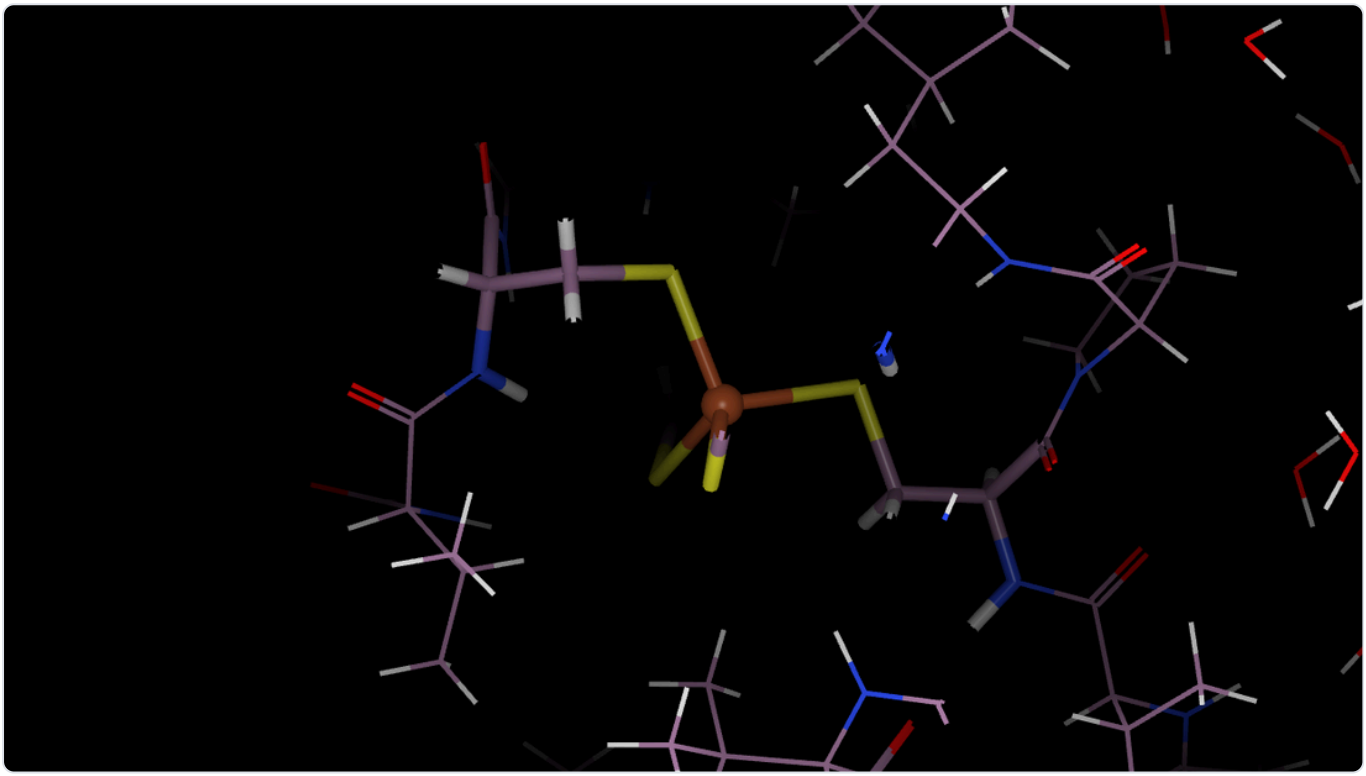


The iron site of 1IRO: SG of Cys42, Cys9, Cys39 and Cys6, 2.23–2.29 Å.

- **Charge and multiplicity of the iron.** The default for Fe is Fe(III), high spin: charge 3, multiplicity 6 (five unpaired d electrons). For the reduced protein use Fe(II), high spin: charge 2, multiplicity 5. The rubredoxin iron is high spin in both oxidation states.
- **Waters** get TIP3P charges and the oxygen Lennard-Jones on DYFF bonded terms. They are not refined with xTB.
- **Charge dialog:** the sum of formal charges is −9 (Fe^{3+}, four thiolates and an acidic protein). Keep −9 and multiplicity 6: the multiplicity goes on the iron site.



The preparation takes about 2 minutes and gives `1IRO_DYFF`: 1112 atoms, total charge −9, the iron bonded to the four SG atoms.



Other metal sites

Case	What to do
Catalytic Zn with a water or hydroxide (carbonic anhydrase, 3KS3)	A coordinated water is left unbonded. For a zinc-bound hydroxide, edit the site in the Builder after the preparation (make the O–Zn bond, remove one H); closing the Builder refines the edited site again with xTB.
Blue copper, Cu(II) $His_2CysMet$ (azurin 4AZU, plastocyanin 1PLC)	Charge 2, multiplicity 2. Choose g-xTB: with GFN2-xTB the weak axial Cu–S(Met) bond comes out too short (2.4 Å instead of 2.8 Å in the crystal).
Mg and Ca (calmodulin 1CLL, thermolysin 8TLN)	Ionic by default: point charge with Lennard-Jones, no bonds. This is the usual model for these hard ions.
Several metals (SOD1 2C9V: Cu and Zn)	Each site has its own row, charge and multiplicity. In the charge dialog, keep the default multiplicity: each site then keeps the one set in its row. A different value is given to the first metal, and the other metals become singlets.

Case	What to do
Sites completed by crystal symmetry (insulin hexamer 4INS)	The asymmetric unit holds only part of the site. It becomes ionic, with a warning. Build the biological assembly first to get the bonded model.
Heme, Fe–S clusters (myoglobin 1A6M, cytochrome c 1HRC, ferredoxin 1FDN)	Not supported yet. The window lists them. Remove them, or build an active-site model in the Builder.

Troubleshooting

Symptom	Cause and fix
"the multiplicity must be odd/even"	The charge and multiplicity do not agree with the number of electrons. With an even number of electrons the multiplicity is odd (1, 3, 5); with an odd number, even (2, 4, 6).
The metal has too few bonds	A donor is farther than the sum of the covalent radii plus 0.5 Å. Check the structure; a distorted NMR model may need a different model of the ensemble.
"atoms could not be given coordinates"	Atom names that do not match the residue templates. Rename them in the PDB file to the standard names.
A system without a force field after the charge dialog	The dialog was cancelled or the multiplicity was refused. Run Prepare DYFF System on the new system again: it already has bonds, so the shorter dialog opens.
Long preparation times	Each metal site and each ligand is one xTB group (1–5 minutes each with GFN2-xTB). g-xTB is 2–3 times slower; the torsion scans add seconds to minutes per flexible bond.

References

- Zinc finger Xfin-31: Lee, M. S.; Gippert, G. P.; Soman, K. V.; Case, D. A.; Wright, P. E. *Science* **1989**, *245*, 635–637.
- Rubredoxin 1IRO: Dauter, Z.; Wilson, K. S.; Sieker, L. C.; Moulis, J. M.; Meyer, J. *Proc. Natl. Acad. Sci. USA* **1996**, *93*, 8836–8840.
- GFN2-xTB: Bannwarth, C.; Ehlert, S.; Grimme, S. *J. Chem. Theory Comput.* **2019**, *15*, 1652–1671.
- Modified Seminario method: Allen, A. E. A.; Payne, M. C.; Cole, D. J. *J. Chem. Theory Comput.* **2018**, *14*, 274–281.
- More in the DYFF manual.

EasyHybrid 3.2.0 · built 2026-10-29 from `dev` @ `3539da4` · https://github.com/ferbachega/EasyHybrid3